Six further OEIS congruence proofs

Expanded proof package, version 3

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AI disclosure and review status. ChatGPT (OpenAI) generated the proofs, mathematical exposition, Lean formalizations, verification programs, and AI reviews for this contribution in response to Dev Goyal's requests. Dev Goyal is the sole human author and initiated and requested the work. The OEIS sequences and conjectures are credited to Paul D. Hanna. **Independent human review is pending.** This paper does not claim OEIS acceptance, peer review, or worldwide priority.

Abstract

We prove six coefficient assertions marked conjectural in the retrieved OEIS entries A397588, A389536, A389537, A389542, A378580, and A378581. The first has odd coefficients exactly at powers of two. Three further series have sparse residue-two support modulo four, and two theta substitution equations have residue-two support exactly at positive squares. Every implicitly defined integer series is constructed and identified by its literal source equation and a uniqueness theorem. The methods are triangular coefficient recursion, genuine compositional inversion, characteristic-two squaring, and controlled passage from modulus two to modulus four. We additionally verify an already stated divisibility-by-three property of A397588. All six target assertions have unbounded Lean proofs; finite calculations independently check their source correspondence. This paper supplements the two earlier papers retained unchanged in the combined archive.

1 Scope, notation, and source status

All series are formal. We write $[x^n]A$ for a coefficient and $A \circ B$ for $A(B(x))$, with the outer series first. Such substitution is used only when $B(0) = 0$. A *normalized* series is $x + O(x^2)$; a *constant-one* series is $1 + O(x)$. The notation $O(x^r)$ means divisibility by x^r, including over rings with zero divisors. Ordinary powers and compositional powers are different operations. The symbol $A^{\langle -1 \rangle}$ denotes compositional inversion, while U^{-1} denotes the multiplicative inverse of a unit. The set $\mathbb{N}$ includes zero.

The official pages were read on 24 October 2026 [1, 2, 3, 4, 5, 6]. Each explicitly called the target in Table 1 a conjecture. Throughout the table, $a_n = [x^n]A$, $k \in \mathbb{N}$, and

$$\Theta(x) = 1 + 2\sum_{j \geq 1} x^{j^2}.$$

The first four series are normalized; the last two have constant coefficient one. In each modulus-four row, all coefficients in the stated range outside the displayed support are zero modulo four.

Entry	Source equation	Target assertion and range
A397588	$A = x + (xA^2)'$	a_n odd iff $n = 2^k$, $n \geq 1$.
A389536	$A(x^2 + 2x^3) = xA$	$a_n \equiv 2 \pmod 4$ iff $n = 2^k + 1$, $n > 1$.
A389537	$A(A - x) = A^2$	$a_n \equiv 2 \pmod 4$ iff $n = 3 \cdot 2^k$, $n > 2$.
A389542	$A(A^2 - x^2) = 4A^3$	$a_n \equiv 2 \pmod 4$ iff $n = 2^k + 1$, $n > 1$.
A378580	$A(x/A) = \Theta$	$a_n \equiv 2 \pmod 4$ iff n is a square, $n > 0$.
A378581	$A(xA) = \Theta$	$a_n \equiv 2 \pmod 4$ iff n is a square, $n > 0$.

Table 1: Six source conjectures addressed in this paper.

The source label records the status of the retrieved page. Exact-identifier searches with *proof* or *conjecture*, and searches for the defining equations, did not identify an earlier proof of these targets. This bounded check does not establish that a result was globally open or that this is its first proof. The standard algebraic ingredients below are not claimed as new discoveries.

One relevant prior result was checked separately. Bhave's proof, recorded in the A196523 comments on 13 October 2026, treats $F = x + xF^{\circ\ell}$ and gives coefficients one modulo $\ell - 1$ [7]. That displayed theorem does not directly state the theta substitution results here; the inspected page contains no linked general theta theorem. This comparison is limited to the inspected statement.

The source's divisibility-by-three assertion for A397588 is included as a proved additional property, not as a seventh conjecture. The sign-run observation in A389537 and the convergence question in A378580 are not addressed. Supporting existence, transport, and coefficient lemmas are proof infrastructure, not additional open problems.

2 Formal-series tools

Lemma 2.1 (Coefficientwise construction)**.** *Let K be a commutative ring. Suppose an operator Φ preserves normalized series and, for $r \geq 2$, satisfies*

$$G - H = O(x^r) \quad \Longrightarrow \quad \Phi(G) - \Phi(H) = O(x^{r+1}).$$

Then Φ has exactly one normalized fixed point in $K[[x]]$. More generally, a residual equation whose degree-r condition determines the next unknown coefficient with multiplier one has a unique solution once its initial coefficients are fixed and its lower-degree conditions hold identically.

Proof. Begin with $G_0 = x$ and set $G_{m+1} = \Phi(G_m)$. Normalization gives $G_1 - G_0 = O(x^2)$. Inductively, $G_{m+1} - G_m = O(x^{m+2})$, so each coefficient eventually stabilizes. Define G using those stable coefficients. The displayed implication ensures that each coefficient of $\Phi(G)$ equals the corresponding stable coefficient, hence $G = \Phi(G)$. If two fixed points agree below degree r, they agree through degree r; induction proves uniqueness.

For the residual version, set the next coefficient to the negative of the known residual term. Its multiplier is one, so the construction uses only ring operations. No earlier equation changes, and every equation is eventually imposed. The same coefficient comparison proves uniqueness. □

Lemma 2.2 (Units and reversion)**.** *Every constant-one series has a unique constant-one multiplicative inverse. Every normalized series has a unique normalized two-sided compositional inverse. These inversions, formal differentiation, and valid substitution commute with coefficient ring homomorphisms.*

Proof. For $UV = 1$ with $U(0) = 1$, the coefficient of degree $n > 0$ is v_n plus a finite expression in earlier coefficients of V, which determines v_n without division. For normalized G, the coefficient of degree $n > 1$ in $G(H)$ is h_n plus an expression in earlier coefficients of H, giving a unique normalized right inverse H.

To see the second inverse identity, construct a normalized right inverse J of H. Associativity gives $G = G \circ (H \circ J) = (G \circ H) \circ J = J$, and hence $H \circ G = x$. The same association proves uniqueness. Each coefficient of a valid composition is a finite polynomial in coefficients. Differentiation multiplies the degree-$(n+1)$ coefficient by $n+1$. Ring homomorphisms therefore preserve these operations and equations; uniqueness identifies the mapped inverses with the inverses of the mapped series. □

Lemma 2.3 (Binary support)**.** *There is a unique normalized integral series C satisfying $C = x + C^2$. Its reduction modulo two is*

$$P(x) = \sum_{k \geq 0} x^{2^k}.$$

More generally, for every integer $d \geq 1$, the unique zero-constant solution of $V + V^2 = x^d$ in $\mathbb{F}_2[[x]]$ is

$$V = P(x^d) = \sum_{k \geq 0} x^{d2^k}.$$

Proof. The operator $G \mapsto x + G^2$ satisfies Lemma 2.1, since $G^2 - H^2 = (G - H)(G + H)$ and $G + H = O(x)$. In characteristic two, every series V satisfies $V(x)^2 = V(x^2)$. This follows by truncating above any fixed degree and applying the polynomial Frobenius identity.

Thus $V + V^2 = x^d$ is a recursion in which the coefficient at an odd positive index is fixed immediately, while the coefficient at an even index n depends on the coefficient at $n/2 < n$. The constant coefficient is fixed to zero. This proves uniqueness. The proposed sum is well-defined coefficientwise because its exponents tend to infinity; adding its square cancels all terms except x^d. Set $d = 1$ to identify the reduction of C. □

Lemma 2.4 (Doubling and substitution)**.** *For $U, V \in (\mathbb{Z}/4\mathbb{Z})[[x]]$,*

$$2U = 2V \quad \Longleftrightarrow \quad U \equiv V \pmod 2.$$

If B, D have zero constant coefficient and $B \equiv D \pmod 2$, then $2H(B) = 2H(D)$ for every series H over $\mathbb{Z}/4\mathbb{Z}$.

Proof. For residues u, v modulo four, $2u = 2v$ exactly when their reductions modulo two agree. Apply this fact to each coefficient. Reduction modulo two commutes with substitution, giving the second assertion. In particular, this lemma never cancels two inside $\mathbb{Z}/4\mathbb{Z}$. □

These arguments are purely formal: there are no analytic convergence assumptions. We now construct each source series before using its modular properties.

3 A397588: a differential equation and binary parity

Theorem 3.1. *There is exactly one normalized $A \in \mathbb{Z}[[x]]$ satisfying*

$$A = x + (xA^2)'. \tag{1}$$

Its coefficients satisfy, for every $n \geq 0$,

$$[x^n]A \text{ is odd} \quad \Longleftrightarrow \quad n = 2^k \text{ for some } k \in \mathbb{N}.$$

Also $A \equiv x \pmod 3$, so $3 \mid [x^n]A$ for every $n > 1$.

Proof. Over any commutative ring, put $\Phi(G) = x+(xG^2)'$. If G is normalized, then $xG^2 = O(x^3)$ and its derivative is $O(x^2)$, so Φ is normalized. For normalized G, H with $G - H = O(x^r)$,

$$x(G^2 - H^2) = x(G - H)(G + H) = O(x^{r+2}).$$

Formal differentiation lowers the order by at most one, including in positive characteristic. Thus $\Phi(G) - \Phi(H) = O(x^{r+1})$. Lemma 2.1 constructs the unique solution over every commutative ring, in particular over $\mathbb{Z}$. Its integer recursion is

$$a_0 = 0, \qquad a_1 = 1, \qquad a_n = (n+1)\sum_{j=1}^{n-1} a_j a_{n-j} \quad (n > 1).$$

This also exhibits integrality directly. Mapping coefficients preserves the equation, and uniqueness identifies the reduction with the corresponding canonical finite-ring solution.

In characteristic two the Leibniz rule gives $(A^2)' = 2AA' = 0$, hence $(xA^2)' = A^2$. Equation (1) becomes $A + A^2 = x$. Lemma 2.3 identifies A modulo two with P, proving the parity criterion. At $n = 0$, both sides of that criterion are false.

In characteristic three, x itself satisfies the equation because $(x^3)' = 3x^2 = 0$. Uniqueness gives $A = x$ in $\mathbb{F}_3[[x]]$, which proves the additional divisibility assertion. □

The parity statement is the conjecture in A397588 [1]. The divisibility by three was already stated there without a conjecture label. It is included here with a proof and a stronger entire-series identity, with no claim that it is an additional newly solved problem.

4 A389536: a Mahler-type substitution modulo four

Theorem 4.1. *The source equation*

$$A(x^2 + 2x^3) = xA(x), \qquad A = x + O(x^2), \tag{2}$$

has a unique solution in $\mathbb{Z}[[x]]$. With $C = x + C^2$ as in Lemma 2.3, it satisfies

$$A(x) \equiv x + 2xC(x) \pmod 4.$$

Consequently, for every $n > 1$,

$$[x^n]A \equiv \begin{cases} 2 \pmod 4, & n = 2^k + 1 \text{ for some } k \in \mathbb{N}, \\ 0 \pmod 4, & \text{otherwise.} \end{cases}$$

Proof. Write $B = x^2 + 2x^3$. For $A = \sum_{n\geq 1} a_n x^n$ with $a_1 = 1$, the coefficient of x^{n+1} in (2), for $n \geq 2$, gives

$$a_n = \sum_{\substack{j\geq 1 \\ 0\leq n+1-2j\leq j}} a_j \binom{j}{n+1-2j} 2^{n+1-2j}. \tag{3}$$

Indeed, $B^j = x^{2j}(1+2x)^j$. Every contributing index obeys $j \leq (n+1)/2 < n$, so this is a finite, division-free recursion in earlier coefficients. The degree-two equation is already satisfied by $a_1 = 1$, and the lower equations vanish. Thus the construction exists and is unique over any commutative ring and commutes with reduction.

Work now modulo four. Denote the reduction of C by the same letter, so $C = x + C^2$, and put $D = x + 2xC$. Its normalization follows from $C(0) = 0$. Modulo two, $B = x^2$, and

Frobenius gives $C(B) = C(x^2) = C^2$. Lemma 2.4 therefore gives $2C(B) = 2C^2$ modulo four. Compute in $(\mathbb{Z}/4\mathbb{Z})[[x]]$:

$$\begin{aligned} D(B) &= B + 2B\,C(B) \\ &= x^2 + 2x^3 + 2x^2 C(B) \\ &= x^2 + 2x^3 + 2x^2 C^2 \\ &= x^2 + 2x^2 C = xD. \end{aligned}$$

The second line discards a multiple of four; the fourth uses $C^2 = C - x$. Thus D satisfies the actual source equation in this ring. Uniqueness identifies it with the reduction of the integer solution.

For $n > 1$, its coefficient is $2[x^{n-1}]C$ modulo four. Lemma 2.3 says this is two precisely when $n - 1 = 2^k$; otherwise it is zero. □

The source writes $A = A(x^2+2x^3)/x$ [2]. Equation (2) is the equivalent division-free identity. Multiplication by x is injective in every formal power-series ring, so this rewriting introduces no additional choice of solution.

5 A389537: an alternating lacunary inverse

5.1 Construction and the original equation

Over any commutative ring, define

$$R(x) = \sum_{k\geq 0} (-1)^k x^{2^k}. \tag{4}$$

This is a normalized formal series, and cancellation of successive terms gives $R(x) + R(x^2) = x$. It is the unique normalized solution of this equation: an odd degree above one has coefficient zero, and an even degree is determined by minus the coefficient at half that degree. Equivalently, $U \mapsto x - U(x^2)$ strictly increases the order of differences between normalized inputs, so Lemma 2.1 applies. In Lean the latter construction is used.

Let $A = R^{\langle -1\rangle}$ be the genuine normalized compositional inverse. Substituting A in $R + R(x^2) = x$ gives

$$x + R(A^2) = A.$$

Apply A to $R(A^2) = A - x$ to obtain the literal A389537 equation

$$A(A(x) - x) = A(x)^2. \tag{5}$$

Conversely, let normalized U satisfy (5), and let $V = U^{\langle -1\rangle}$. Applying V on the outside gives $U - x = V(U^2)$. Substituting V for x then gives $x - V = V(x^2)$. The uniqueness of (4) implies $V = R$ and hence $U = A$. We have therefore constructed the unique normalized integral solution of the original equation, not just an auxiliary series with a related congruence. Coefficient maps preserve this construction by Lemma 2.2.

5.2 Parity, exact halving, and the modulus-four result

Theorem 5.1. *For the preceding integer source series, $a_1 = a_2 = 1$, and for every $n > 2$,*

$$a_n \equiv \begin{cases} 2 \pmod 4, & n = 3 \cdot 2^k \text{ for some } k \in \mathbb{N}, \\ 0 \pmod 4, & \text{otherwise.} \end{cases}$$

Proof. Modulo two, Frobenius changes the inverse equation to $R+R^2 = x$. Thus $(x+x^2)\circ R = x$. The uniqueness of normalized compositional inverses shows

$$A \equiv x + x^2 \pmod 2. \tag{6}$$

In particular, there is a well-defined integral series

$$H = \frac{A - x - x^2}{2} \in \mathbb{Z}[[x]], \qquad H(0) = 0. \tag{7}$$

This division takes place coefficientwise in the integers, after the divisibility proof (6); it is not division by two inside $\mathbb{Z}/4\mathbb{Z}$. The initial values $a_1 = a_2 = 1$ also follow directly from $R = x - x^2 + O(x^4)$ and $R(A) = x$.

Reduce (7) modulo four, retaining the notation H, and set $D = A - x = x^2 + 2H$. Both D and x^2 have zero constant coefficient, and they have the same reduction modulo two. Lemma 2.4 therefore gives $2H(D) = 2H(x^2)$. Expanding the source equation yields

$$\begin{aligned} A(D) &= D + D^2 + 2H(D) = x^2 + 2H + x^4 + 2H(x^2), \\ A^2 &= (x + x^2 + 2H)^2 = x^2 + 2x^3 + x^4. \end{aligned}$$

All equalities in this calculation are in $(\mathbb{Z}/4\mathbb{Z})[[x]]$. After subtraction,

$$2\big(H + H(x^2)\big) = 2x^3.$$

Applying the equivalence in Lemma 2.4, and writing $\overline{H}$ for reduction modulo two, gives

$$\overline{H} + \overline{H}(x^2) = x^3, \qquad \text{equivalently} \qquad \overline{H} + \overline{H}^{\,2} = x^3.$$

Since $\overline{H}(0) = 0$, Lemma 2.3 identifies it uniquely:

$$\overline{H} = \sum_{k \geq 0} x^{3 \cdot 2^k}.$$

For $n > 2$, equation (7) gives $a_n \equiv 2[x^n]H \pmod 4$. The displayed support proves both residue cases. □

This is the modulus-four conjecture in A389537 [3]. The same entry's empirical observation about lengths of sign runs is a separate problem and is not a consequence claimed here.

6 A389542: an integral inverse that becomes an involution

6.1 An integral construction and exact source correspondence

Construct a normalized series L over any commutative ring from

$$L = x + L^2 - 2xL(4x^3). \tag{8}$$

The right side preserves normalization. If $G - H = O(x^r)$, then $G^2 - H^2 = O(x^{r+1})$ for normalized inputs, while $x(G - H)(4x^3) = O(x^{3r+1})$. Lemma 2.1 proves existence and uniqueness without division. More explicitly, for $n \geq 2$,

$$\ell_n = \sum_{i=1}^{n-1} \ell_i \ell_{n-i} - \begin{cases} 2\,4^{(n-1)/3}\ell_{(n-1)/3}, & 3 \mid n-1, \\ 0, & 3 \nmid n-1, \end{cases} \qquad \ell_1 = 1.$$

Set $R = x - 2xL$. Equation (8) gives

$$\begin{aligned} R^2 + R(4x^3) &= x^2 + 4x^2(-L + L^2 + x - 2xL(4x^3)) \\ &= x^2. \end{aligned} \tag{9}$$

Let $A = R^{\langle -1 \rangle}$, which exists integrally by Lemma 2.2. Composing (9) on the right with A gives $x^2 + R(4A^3) = A^2$. Applying A to $R(4A^3) = A^2 - x^2$ yields the exact source equation

$$A(A(x)^2 - x^2) = 4A(x)^3. \tag{10}$$

For completeness, this normalized integer source solution is unique. If a normalized $U \in \mathbb{Z}[[x]]$ satisfies (10) and $V = U^{\langle -1 \rangle}$, substitution of $V(t)$ gives $U(t^2 - V(t)^2) = 4t^3$. Applying V yields

$$V(t)^2 + V(4t^3) = t^2. \tag{11}$$

If two normalized integer solutions of (11) first differ by $cx^r + O(x^{r+1})$, $r \geq 2$, their squares first differ by $2cx^{r+1} + O(x^{r+2})$. Their substituted difference has order at least $3r \geq r + 2$. Subtracting the equations forces $2c = 0$ in $\mathbb{Z}$, and hence $c = 0$, a contradiction. Thus $V = R$ and $U = A$. The uniqueness argument here is specifically over $\mathbb{Z}$; cancellation of two is not asserted over rings with two-torsion.

6.2 Reduction modulo four

Theorem 6.1. *The unique normalized integer solution of* (10) *satisfies, for every* $n > 1$,

$$[x^n]A \equiv \begin{cases} 2 \pmod 4, & n = 2^k + 1 \text{ for some } k \in \mathbb{N}, \\ 0 \pmod 4, & \text{otherwise.} \end{cases}$$

Proof. The construction of L works uniquely over any commutative ring. It therefore commutes with reduction, as do $R = x - 2xL$ and its genuine compositional inverse. Modulo two, equation (8) becomes $L + L^2 = x$. Lemma 2.3 identifies its support with the powers of two.

Now work modulo four. Since $R = x - 2xL \equiv x \pmod 2$, substitution gives $L(R) \equiv L \pmod 2$, and Lemma 2.4 gives $2L(R) = 2L$. Consequently

$$\begin{aligned} R(R) &= R - 2R\,L(R) = R - 2R\,L \\ &= (x - 2xL)(1 - 2L) = x - 4xL + 4xL^2 = x. \end{aligned}$$

Thus R is its own compositional inverse modulo four. Uniqueness of compositional inverses identifies the reduction of A with R:

$$A \equiv x - 2xL \pmod 4.$$

For $n > 1$, the coefficient is $-2\ell_{n-1}$ modulo four. This equals two precisely when ℓ_{n-1} is odd, which holds exactly when $n - 1 = 2^k$; otherwise it is zero. □

The result is the single modulus-four conjecture in A389542 [4]. The entire involution argument uses multiplication by two and reduction to characteristic two; it makes no cancellation of two in $\mathbb{Z}/4\mathbb{Z}$.

7 A378580 and A378581: theta substitution

7.1 A general integral construction

The two source equations are

$$A_-(x/A_-(x)) = \Theta(x), \qquad A_+(xA_+(x)) = \Theta(x), \qquad A_\pm(0) = 1. \tag{12}$$

We first solve them with an arbitrary constant-one target T in place of Θ, over any commutative coefficient ring K.

Lemma 7.1. *For each $T \in K[[x]]$ with $T(0) = 1$, there is exactly one normalized G satisfying*

$$G \circ G = GT. \tag{13}$$

Proof. Let $E(G) = G \circ G - GT$. Its coefficients in degrees zero and one vanish for every normalized G. Suppose $G - H = dx^r + O(x^{r+1})$, with $r \geq 2$. Decompose the difference of self-compositions as

$$G \circ G - H \circ H = (G - H) \circ G + (H \circ G - H \circ H).$$

Because $G = x + O(x^2)$, the first summand is $dx^r + O(x^{r+1})$. Write $H = x + x^2W$. The second summand is $(G - H) + G^2W(G) - H^2W(H)$. The difference after $G - H$ has order at least $r + 1$: factor $G^2 - H^2$, and use that substitution in any series preserves an agreement modulo x^r. Thus the second summand also is $dx^r + O(x^{r+1})$.

Finally, $(G - H)T = dx^r + O(x^{r+1})$, since $T(0) = 1$. Therefore

$$E(G) - E(H) = dx^r + O(x^{r+1}).$$

The degree-r residual has multiplier $2 - 1 = 1$ on the next unknown coefficient. Lemma 2.1's triangular construction proves existence and uniqueness with no division, over the stated ring. □

Write this unique G as $G = xU$. Then $U(0) = 1$ and

$$G \circ G = G\,U(G) = GT.$$

Cancelling G is valid even when K has zero divisors: multiplication by x is injective coefficient-twise, and U is a unit. It follows that $U(xU) = T$. Conversely, any constant-one solution of $U(xU) = T$ gives a normalized solution xU of (13). Consequently there exists exactly one such U; denote it by $\mathrm{Plus}(T)$.

Define

$$\mathrm{Minus}(T) = \mathrm{Plus}(T^{-1})^{-1},$$

where both occurrences of inverse here are multiplicative unit inverses. If $U = \mathrm{Plus}(T^{-1})$, then $U(xU) = T^{-1}$. Substitution respects products, so $U^{-1}(xU) = T$. Since $x/\,\mathrm{Minus}(T) = xU$, this is

$$\mathrm{Minus}(T)(x/\,\mathrm{Minus}(T)) = T.$$

Conversely, if constant-one V satisfies $V(x/V) = T$, then inversion of that identity gives $V^{-1}(xV^{-1}) = T^{-1}$. The uniqueness of $\mathrm{Plus}(T^{-1})$ identifies $V^{-1} = U$ and hence $V = \mathrm{Minus}(T)$. We have proved existence and uniqueness for both source equations over every commutative ring, including $\mathbb{Z}$.

The quotient is literal: $V(x)\,(xV(x)^{-1}) = x$. It is not interpreted as a compositional inverse. Coefficient maps preserve the two source equations and their unit inverses, so uniqueness also proves that Plus and Minus commute with reduction of coefficients.

7.2 The common square-support congruence

Theorem 7.2. *Let* $A_- = \mathrm{Minus}(\Theta)$ *and* $A_+ = \mathrm{Plus}(\Theta)$ *over* $\mathbb{Z}$. *These are the unique constant-one integral solutions of* (12), *and*

$$A_-(x) \equiv A_+(x) \equiv \Theta(x) \pmod 4.$$

In particular, for either sign and every $n > 0$,

$$[x^n]A_\pm \equiv \begin{cases} 2 \pmod 4, & n = j^2 \text{ for some } j \in \mathbb{N}, \\ 0 \pmod 4, & \text{otherwise.} \end{cases}$$

Proof. Work over $\mathbb{Z}/4\mathbb{Z}$ and put $S = \sum_{j\geq 1} x^{j^2}$, so $\Theta = 1 + 2S$. The series $B = x\Theta$ has zero constant coefficient and satisfies $B \equiv x \pmod 2$. Lemma 2.4 gives $2S(B) = 2S(x)$. Therefore

$$\Theta(x\Theta(x)) = 1 + 2S(x\Theta(x)) = 1 + 2S(x) = \Theta(x).$$

The uniqueness just proved identifies $\mathrm{Plus}(\Theta) = \Theta$ over this ring. Also

$$\Theta^2 = (1 + 2S)^2 = 1 + 4S + 4S^2 = 1,$$

so $\Theta^{-1} = \Theta$. It follows that

$$\mathrm{Minus}(\Theta) = \mathrm{Plus}(\Theta^{-1})^{-1} = \mathrm{Plus}(\Theta)^{-1} = \Theta^{-1} = \Theta.$$

Compatibility with coefficient reduction identifies these series with the reductions of the canonical integer solutions. Finally, the coefficient of Θ at a positive index is two exactly when the index is a square, and zero otherwise. □

This proves one original coefficient conjecture in each of A378580 and A378581 [5, 6]. Each equivalence is accompanied by the complementary integer statement $4 \mid a_n$ at nonsquares. The constant term is one in both sequences; excluding $n = 0$ is essential. The modular argument only uses the form $1 + 2S$ and $S(0) = 0$, so its mechanism is more general than square support. No additional specializations are counted as new problems in this paper.

8 Formal coverage, corroboration, and reproduction

The accompanying project uses Lean 4.19.0 and Mathlib revision

`c44e0c8ee63ca166450922a373c7409c5d26b00b.`

The mathematical objects are actual Mathlib formal power series, derivatives, substitutions, and coefficient homomorphisms. The modules under `lean/Round3/` implement the arguments as follows:

Module	Formal content
`DifferentialParity`	Integral source solution, uniqueness, binary parity, and the additional modulus-three property.
`MahlerParity`	Literal source solution, uniqueness, comparison with $x+2xC$, and both residue clauses.
`LacunaryInverse`	Auxiliary construction, genuine inverse, both source-equation directions, exact integer halving, and support.
`SquareReversion`	Integral auxiliary solution, source identification and uniqueness, involution modulo four, and residues.
`ThetaSubstitution`	General constant-one source solutions, literal unit quotient, uniqueness, reduction, and both square criteria.
`ModFour`	Exact scalar and series doubling lemmas and justified coefficientwise integer halving.

Separate typed contract files restate the literal equations, coefficient domains, initial conditions, uniqueness claims, and unrestricted index ranges. In particular, the quotient contract uses `PowerSeries.invOfUnit` explicitly; reversion contracts use proved two-sided compositional inverses. Thus the target sequences are not replaced by unrelated series defined from the desired residue pattern.

The package's validation process rebuilds the first-party proof modules, compiles the contracts, prints fully elaborated endpoint types, and audits their transitive axiom dependencies. The permitted foundational axioms are `propext`, `Classical.choice`, and `Quot.sound`. The proof sources use no `sorry`, `admit`, `native_decide`, or custom mathematical axiom. Finite identities among residues are proved with kernel-checked `decide`; they are ingredients in all-index series proofs, not substitutes for them. The delivered validation report records the actual run and its endpoint list. Endpoint counts measure proof coverage, not counts of open problems.

AI reviewers additionally examined the connection between the formal definitions and the source statements. These semantic reviews are distinct from kernel checking, and neither establishes historical novelty or human endorsement. Independent human review remains pending.

Independent finite corroboration. Exact-integer programs construct coefficients from the original equations, with separate auxiliary constructions where applicable. They match all displayed source terms and check the following finite ranges:

Entries	Recorded finite checks
A397588	Differential identity and parity through degree 500; divisibility by three from degree 2 through 500.
A389536	Residue pattern and Catalan comparison through degree 2050; direct generic substitution through degree 81.
A389537	Direct source recursion agrees with reversion of the alternating lacunary series through degree 80.
A389542	Direct source recursion agrees with the auxiliary construction through degree 80; source and inverse equations checked through degree 81.
A378580–581	Direct source recursions agree with the separate normalized-root construction through degree 64; both source equations and residue clauses checked.

These finite ranges are evidence about transcription and source identification. The coefficient conclusions proved in Lean are unbounded. The machine-readable evidence is under `validation/`; individual proof notes explain the exact checks and source prefixes.

Reproduction and retained work. From the extracted package, with the pinned Lean toolchain and Python installed, run:

```
cd lean
lake exe cache get
lake build
cd ..
python3 scripts/check_formalization.py
```

The project records dependency pins, explicit acceptance contracts, source hashes, build logs, theorem types, and axiom-audit results. The package-wide report is authoritative for the particular delivered run.

The combined archive retains the two earlier papers and their proof modules unchanged. This paper adds the six assertions in Table 1 and the separately identified additional property. The user-supplied repository and its author details are credited [8]; its earlier identifier or DOI is not represented as a DOI for this expanded work. No submission to OEIS or publication acceptance is claimed.

References

[1] P. D. Hanna, OEIS A397588, 3 July 2026. `https://oeis.org/A397588`. Accessed 24 October 2026.

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[8] D. Goyal, *oeis-a396843-proof*, user-supplied repository. `https://github.com/dvgyl/oeis-a396843-proof`. Original source revision and attribution are preserved in the package.

[9] Mathlib contributors, formal proof library for Lean. `https://github.com/leanprover-community/mathlib4`. The exact dependency revision is pinned in `lean/lake-manifest.json`.